to obtain an access token with any scope it wants. This is almost identical to the token login endpoint you developed in previous chapters and is not recommended for third-party clients because the user directly shares their password with the app—the very thing you were trying to avoid!

> **CAUTION** ROPC can be useful for testing but should be avoided in most cases.

- In the *Authorization Code* grant, the client first uses a web browser to navigate to a dedicated *authorization endpoint* on the AS, indicating which scopes it requires. The AS then authenticates the user directly in the browser and asks for consent for the client access. If the user agrees then the AS generates an authorization code and gives it to the client to exchange for an access token at the token endpoint. The authorization code grant is covered in more detail in the next section.
- The *Client Credentials* grant allows the client to obtain an access token using its own credentials, with no user involved at all. This grant can be useful in some microservice communications patterns discussed in part 3 of this book, but it's often better to use a different grant if you can.
- There are several additional grant types for more specific situations, such as the *device flow grant* for devices without any direct means of user interaction. There is no registry of defined grant types, but websites such as https://oauth.net/2/grant-types/ list the most commonly used types. The device flow is covered in part 4 on the Internet of Things. OAuth2